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Subject: Engineering Physics - IA 1 | Section: A

1.

First law - an object at rest stays at rest and an object in motion keeps moving with the same velocity unless a force acts on it (inertia).

Second law - force is equal to mass times acceleration, F = ma.

Third law - every action has an equal and opposite reaction. Example: a rocket pushes gases down and the gases push the rocket up.

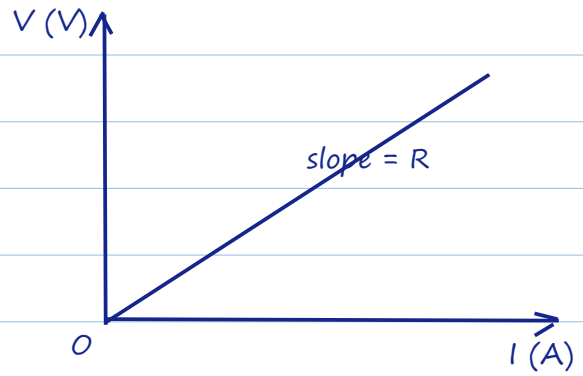
2.

a = (20 - 0)/5 = 4 m/s^2

distance s = v x t = 20 x 5 = 100 m

3.

Ohm's law states that V is directly proportional to I, V = IR, when temperature is constant.



R is found from the slope of the graph.

4.

Kinetic energy = energy of a moving body = 1/2 mv^2

Potential energy = energy stored due to height = mgh

PE at top = 2 x 10 x 20 = 400 J. All of it becomes KE at the bottom.

400 = 1/2 x 2 x v^2 so v^2 = 400, v = 20 m/s